

Studio 9

Assignment 3 Framework preparation

Objectives

To build the framework for assignment 3.

Exercise 1

You will be building a framework to allow the user to choose different visualization of data to view. The dataset stores the id number, internal mark and exam mark for students. The data set will visualize the data by displaying a scatter plot of internal mark vs exam mark. The internal mark is along the x axis and the exam mark is along the y axis.

1. Download the template project and open it in Processing. Open the class diagram.
2. Implement the StudentMarks class using the notes below:

The constructor will initialise the object using the values passed in.

The `getInternal()` and `getExam()` methods will return back the internal mark and exam mark respectively.

The `PadRight` method is a private method that will pad the given string with spaces so the number of characters matches the width value passed in. The method then returns back the padded string.

The `plotMark()` method will draw a small circle in the correct colour at the correct x and y position. You need to work out its x and y position from the internal and exam mark. The marks are out of 100.

The `toString()` method will return back the id number, internal mark and exam mark as a neatly padded out string.

Test your class by uncommenting the testing code in the sketch tab then commenting out the code again when it is correct.

3. Implement the DataSet class using the notes below:

The constructor will set the data set colour to the colour value passed in. It will also create the new ArrayList object. Then it will load the csv file of data using the filename passed into the method. The format of the csv file is: `id, internal mark, exam mark`

The `plotDataSet()` method will plot all the marks as a scatter plot without a stroke.

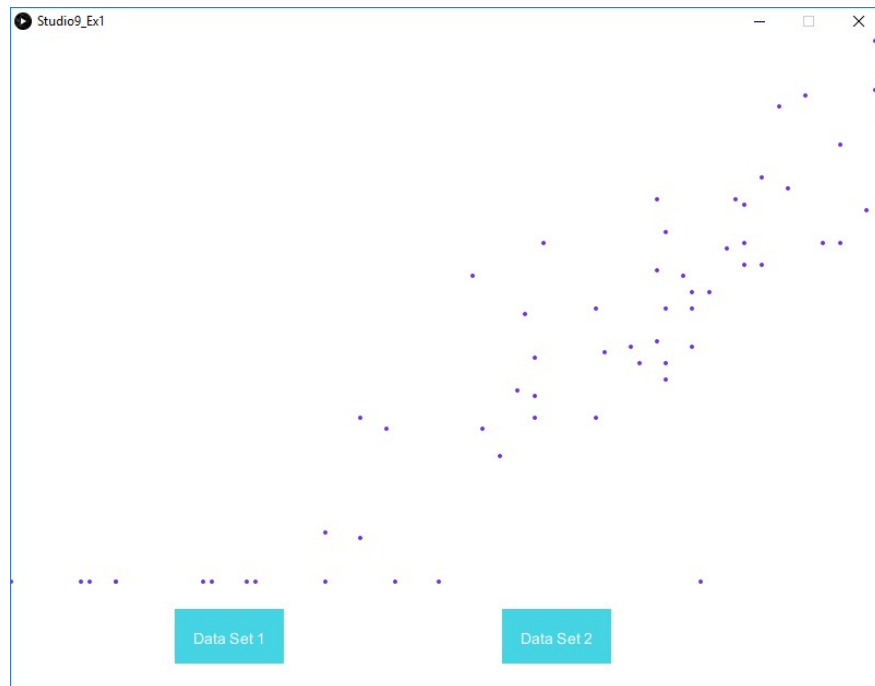
The `displayDataSet()` method will display all the data in the dataset to the console window. Note: just print the values to the console window in this method rather than returning everything as one big string.

The `getAverageInternal()` and `getAverageExam()` methods will return back the average internal mark and average exam mark respectively.

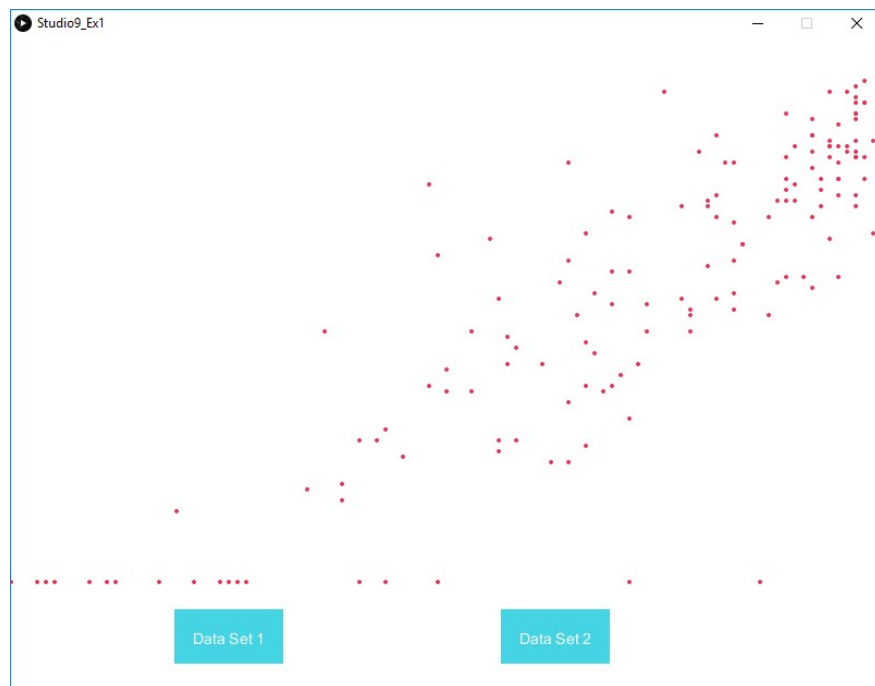
Test your class by uncommenting the testing code in the sketch tab.

Exercise 2 (Studio Handin Exercise)

1. Add your button class from previous studios to the sketch which allows you to create rectangular buttons and store a caption for the button.
2. Create two buttons for `DataSet1` and `DataSet2` and give them appropriate coordinates so that they are somewhere near the bottom of the sketch window.
3. Create the `mousePressed()` method which checks if either of the two buttons is clicked on. If one of them is clicked on then display the appropriate data set.
4. Figure out how to change the code so that the bottom y position of the graph is passed to the `plotDataSet()` method and it is passed on to the `plotMark()` method and used appropriately.
5. Now change the sketch tab code so that the bottom of the graph is just above your buttons so that the buttons don't interfere with the graph. See the screen shots below:



Data Set 1 screenshot



Data Set 2 screenshot